Experiment 4: The Sol-Gel Technique – Lab Report

Name: Yunus Yener **Student No:** 2205C046 **Group-5**

1) Aim

To synthesize SiO_2 nanopowder by the sol-gel method, coat it by dip and spin coating, and examine the effect of SiO_2 content on the contact angle and viscosity of PVA solutions.

2) Materials and Devices

TEOS, ethanol, distilled water, NH_4F, NH_4OH, PVA, Triton X-100; magnetic stirrer, balance, drying oven, furnace, laser, spin coater, FTO/ITO glass, Ostwald viscometer.

3) Experimental Procedure

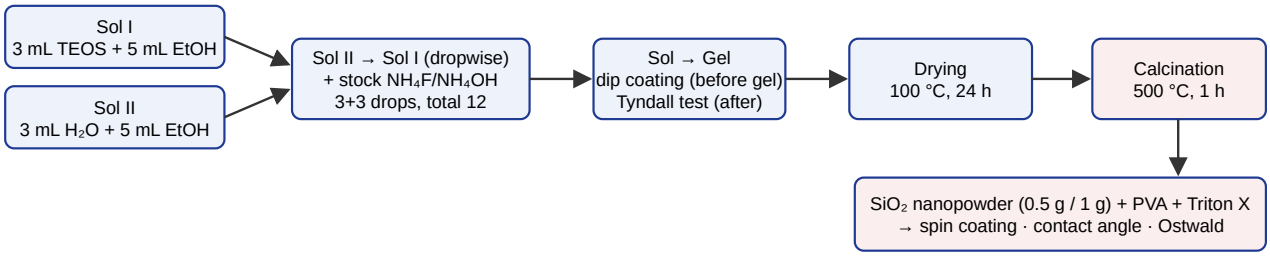


Figure 1. Flow chart of the experiment.

- Solution II (3 mL water + 5 mL ethanol) was added dropwise to Solution I (3 mL TEOS + 5 mL ethanol) under stirring; hydrolysis started (≡Si–OR + H_2O → ≡Si–OH + ROH).
- Stock catalyst (1.9 g NH_4F + 23 mL NH_4OH in 100 mL water [1]) was added 3 drops at a time until a coating formed on a dipped glass slide (12 drops in total).
- Before gelation a glass slide was dipped into the sol and slowly withdrawn (dip coating). After gelation a laser beam was passed through a beaker of water and then through the gel.
- The wet gel was weighed, dried at 100 °C for 24 h, calcined at 500 °C for 1 h and weighed after each step to obtain the SiO_2 nanopowder.
- PVA suspensions with 0.5 g and 1 g SiO_2 were spin-coated (3 drops, 500/800 rpm, 5/10 s). One drop of water was placed on each coating, and the viscosities were measured with the Ostwald viscometer (5 mL, water as reference).

4) Results and Discussion

4.1) Percent weight loss

Step	Mass (g)	Loss (%)
Wet gel	10.900	–
Dried (100 °C, 24 h)	0.576	94.72
Calcined (500 °C, 1 h)	0.394	31.60
Total		**96.39**

$$Loss\ (\%) = (m_{before} - m_{after}) / m_{before} \times 100$$

Drying: (10.9 − 0.576)/10.9 × 100 = 94.72 %
Calcination: (0.576 − 0.394)/0.576 × 100 = 31.60 %
Total: (10.9 − 0.394)/10.9 × 100 = 96.39 %

Most of the mass was lost during drying, because the wet gel is mostly water and ethanol trapped in the silica network. During calcination, residual organic groups burned off and Si–OH groups condensed into Si–O–Si.

4.2) Tyndall effect

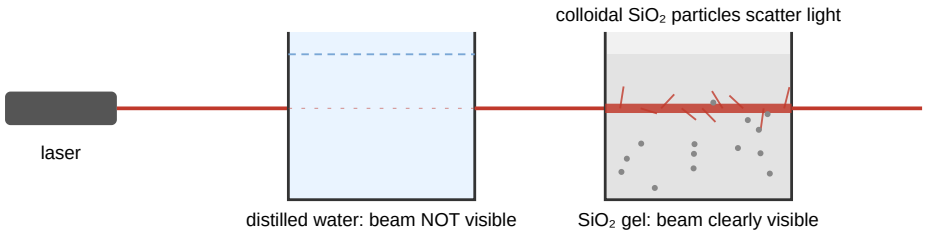
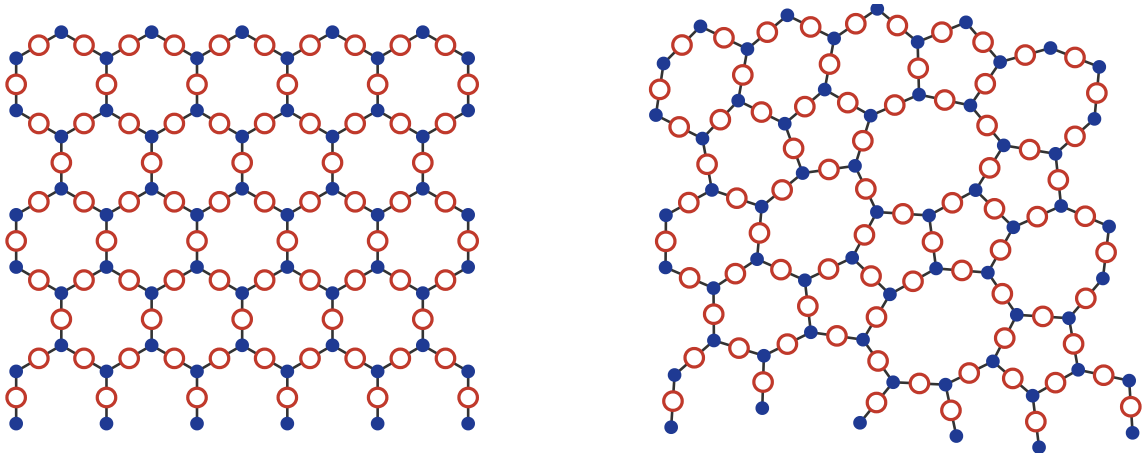


Figure 2. Observation of the Tyndall effect.

The beam was invisible in water but clearly visible in the gel. The colloidal SiO_2 particles, close in size to the wavelength of light, scatter the beam, while a true solution does not. This shows that the gel is a colloidal system.

4.3) Crystalline and amorphous SiO_2



(a) Crystalline SiO_2 *(b) Amorphous SiO_2*

● Si ○ O *Figure 3. 2-D schematic of the SiO_4 network.*

Property	Crystalline (quartz)	Amorphous (vitreous)
Order	Long-range, periodic	Only short-range, random rings
Density	2.65 g/cm³	~2.2 g/cm³
Melting	Sharp melting point	Softens gradually (T_g)
XRD / optics	Sharp peaks, anisotropic	Broad hump, isotropic
Chemical	Few Si–OH, less reactive	Many Si–OH, high surface area, more reactive

Both structures are built from SiO_4 tetrahedra; only their arrangement differs. **Our powder is amorphous**, because it formed by random condensation and was heated only to 500 °C, far below the crystallization temperature. **Applications:** fillers for polymers and rubber, catalyst supports, adsorbents (silica gel), drug delivery, water-repellent coatings and polishing slurries.

4.4) Contact angle and superhydrophobic surface

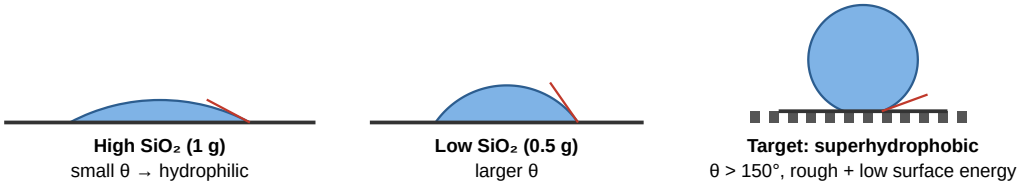


Figure 4. Observed drop shapes (schematic) and the target surface.

θ < 90° means hydrophilic, θ > 90° hydrophobic and θ > 150° superhydrophobic. **As SiO_2 increased, the contact angle decreased and the surface became more hydrophilic**, because silica carries polar Si–OH groups that form hydrogen bonds with water. **Chemical treatment:** the Si–OH groups must be replaced by low-surface-energy groups by silanization with a fluoroalkylsilane (FAS) [2]:

$$\equiv Si–OH + (C_2H_5O)_3Si–CH_2CH_2(CF_2)_nCF_3 \rightarrow \equiv Si–O–Si–CH_2CH_2(CF_2)_nCF_3 + C_2H_5OH$$

The fluorinated groups lower the surface energy, and the SiO_2 nanoparticles give roughness that traps air under the drop. Brassard et al. spin-coated FAS-treated silica nanoparticles and obtained a contact angle of 151°, so rain drops roll off [2].

4.5) Spin coating: effect of rpm and time

Higher rpm → thinner film: the final thickness follows $h \propto \omega^{-1/2}$ [3], so going from 500 to 800 rpm gives a ~20 % thinner film. **Longer time → thinner only in the first seconds:** after that, solvent evaporation makes the film too viscous to flow and the thickness stays constant [3]. This is why 5 s and 10 s gave the same coating, and rpm was the controlling parameter.

4.6) Viscosity

Liquid	ρ (g/cm³)	t (s)	η (mPa·s)

$$\eta = \eta_0 \cdot (\rho \cdot t) / (\rho_0 \cdot t_0)$$

Water (ref.)	1.000	1.45	0.89
Low SiO_2	1.02586	8.06	**5.08**
High SiO_2	1.02664	8.34	**5.26**

ρ = m/V: 5.1293/5 = 1.02586 ; 5.1332/5 = 1.02664 g/cm³
Low: 0.89 × (1.02586 × 8.06)/(1 × 1.45) = 5.08 mPa·s
High: 0.89 × (1.02664 × 8.34)/(1 × 1.45) = 5.26 mPa·s

The viscosity increased from 5.08 to 5.26 mPa·s as the SiO_2 content increased, because more particles increase internal friction and their Si–OH groups form hydrogen bonds with PVA.

5) Conclusion

Amorphous SiO_2 nanopowder was synthesized by the sol-gel method (total weight loss 96.39 %). More SiO_2 made the coating more hydrophilic and the PVA solution more viscous. Spin speed, not time, controls film thickness, and FAS silanization can make the silica surface superhydrophobic.

6) References

1. Experiment 4: The Sol-Gel Technique, MSE4121 Material Production Laboratory Experimental Booklet, Yıldız Technical University, 2026.
2. Brassard, J.-D., Sarkar, D. K., Perron, J., "Synthesis of monodisperse fluorinated silica nanoparticles and their superhydrophobic thin films", *ACS Appl. Mater. Interfaces* 3 (2011) 3583–3588. doi:10.1021/am2007917
3. Meyerhofer, D., "Characteristics of resist films produced by spinning", *J. Appl. Phys.* 49 (1978) 3993–3997. doi:10.1063/1.325357